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Characterization of *Colletotrichum siamense* Causing Leaf Anthracnose on *Cornus officinalis* and Its In Vitro Sensitivity to Fungicides in China

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Abstract

Cornus officinalis is a valuable traditional Chinese medicinal (TCM) plant species with both therapeutic and ornamental attributes. It is widely used in TCM prescriptions to nourish the liver and kidneys and constitutes a critical component of numerous classical formulas. In recent years, the large-scale cultivation of this medicinal plant has been expanded in Xixia County, Henan Province, China. Field investigations have revealed widespread brown leaf spot, accompanied by reductions in yield and quality. In this study, symptomatic leaves were collected for pathogen isolation. Tissue isolations consistently yielded a *Colletotrichum* fungus, and morphology combined with multi-locus phylogenetic analyses (the internal transcribed spacer, glyceraldehyde-3-phosphate dehydrogenase, chitin synthase, actin, and β-tubulin) identified the pathogen as *Colletotrichum siamense*. Pathogenicity assays (conducted by either wounding and inoculating detached leaves with a mycelium plug or spraying a conidium suspension on healthy potted plants) reproduced field symptoms, and the pathogen was re-isolated, thereby fulfilling Koch's postulates. In vitro fungicide assays showed that carbendazim, tebuconazole, and prochloraz were highly effective against the pathogen, providing preliminary information for chemical management. This is the first documentation of *C. siamense* causing leaf anthracnose on *C. officinalis* and provides a basis for developing targeted control strategies to mitigate disease impacts and preserve yield and quality.

Keywords: *Cornus officinalis*; *Colletotrichum*; anthracnose; medicinal tree; multi-locus phylogenetic analysis

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1. Introduction

Cornus officinalis Sieb. et Zucc., a member of the Cornaceae family, is a small shrub native to East Asia, with major populations in China, Korea, and Japan. It is a traditional Chinese medicinal plant, and its dried, ripened fruit is widely recognized as a functional food and processed into value-added products such as beverages, vinegar, and wine [1]. Harvesting typically occurs from September to October, although fruits are sometimes collected in late autumn or early winter [2]. In traditional Chinese medicine (TCM), the fruit is characterized as sour, astringent, and warming, and is reputed to nourish the liver and kidneys, reduce excessive perspiration, and ameliorate age-associated disorders such as Alzheimer's disease [3]. It plays a key role in numerous classical prescriptions [4]. Clinically, it ranks among the most extensively utilized herbal medicines worldwide and

has applications in pharmaceuticals, food, and cosmetic formulations owing to its anti-inflammatory, antiviral, and antioxidant properties [5–7].

Colletotrichum is among the most prevalent plant-pathogenic fungi worldwide, capable of infecting a broad spectrum of commercially important woody and herbaceous hosts [8–13]. Numerous species within this genus have been documented to infect medicinal plants, as well as numerous herbaceous and woody species, causing anthracnose, which leads to significant yield reductions and economic losses [14,15]. Currently, 16 species complexes are recognized within the genus *Colletotrichum*. Among these, the *C. gloeosporioides* complex contains the most species, followed by the *C. acutatum* and *C. boninense* complexes [16]. *Colletotrichum siamense*, a member of the *C. gloeosporioides* species complex, has previously been reported as a pathogen of medicinal plants such as *Capsicum annuum* and *Morus alba* in China [17,18]. *C. siamense* and *C. gloeosporioides* are associated with the greatest number of host species in the genus, with 1358 host species recorded for *Colletotrichum* [15]. These observations align with the tendency of *C. siamense* populations to occur predominantly in tropical and subtropical climates. This fungus is recognized as a common pathogen with an exceptionally broad host range.

C. officinalis is an important medicinal woody crop widely cultivated in China. Xixia County in Henan Province represents a core production region, with approximately 220,000 mu under cultivation and accounting for more than half of the national output, highlighting the economic importance of managing destructive foliar diseases in commercial orchards. Several diseases have been reported on *C. officinalis*, including leaf blight caused by *Didymella glomerata* [19], leaf blight associated with *Botryosphaeria dothidea* [20].

Systematic disease monitoring was conducted in Xixia County during the summers of 2023 and 2024 to support pathogen management programs for this species. Field inspections revealed numerous severely affected leaves displaying symptoms such as brown lesions, marginal scorch, and tip necrosis. To characterize the disease etiology, we performed isolations, followed by morphological examination and multi-locus phylogenetics of fungal isolates to achieve species-level identification. Moreover, we assessed the pathogenicity of isolates to fulfill Koch's postulates.

2. Materials and Methods

2.1. Sample Collection

Xixia County is one of the most important *C. officinalis* estate regions in the country, with a cultivation area of about 13,000 hectares. Samples were collected in Taiping Town (33°37′ N, 111°43′ E) and Miping Town (33°34′ N, 111°24′ E), Xixia County, Henan Province, during the summers of 2023 and 2024. Environmental context was compiled from Taiping Town and Miping Town for June to September during 2023 and 2024. We summarized daily mean/max/min temperature, relative humidity, and rainfall, and noted irrigation regime, pruning status, planting density, and cultivar at each site. A five-point sampling strategy was implemented within each orchard, encompassing a total of 50 trees. From each tree, five leaves exhibiting leaf spot symptoms were randomly collected. At each study location, samples were taken from 10 trees. Individual samples were placed into separate paper bags, labeled, and promptly transported to the laboratory for fungal isolation and subsequent DNA extraction.

2.2. Isolation and Microscopy

A total of 80 specimens from randomly selected leaves were examined microscopically, and the incidence of disease in the field was subsequently determined. Samples were processed as follows: symptomatic leaves were photographed and then pretreated via rinsing under running water for 10 min to remove surface debris. Leaf surfaces were

disinfected with 75% ethanol for 30 s, treated with 1% NaClO for 60 s, and rinsed with sterile water, followed by drying with sterile absorbent paper. The pretreated samples were transferred to a laminar flow cabinet, and 5 × 5 mm^2 tissue sections were excised from the transition area between the healthy and symptomatic leaf and then inoculated onto potato dextrose agar (PDA) prepared with 200 g of potato, 20 g of glucose, and 15 g of agar per liter and supplemented with streptomycin at 50 μg mL^{-1} (3–5 blocks per plate). Plates were incubated in an inverted position at 25 °C.

Mycelia from the colony margin were sampled using a 3 mm punch and inoculated at the center of PDA plates for three days to observe colony morphology, color, and size, with photographs taken of individual colonies. After seven days of incubation, the mycelium from the colony margin was used to prepare conidium suspensions with Triton X-100 reagent. Morphological assessments were conducted on 10-day-old cultures using a microscope (BM2100, Jiangnan, Suzhou, China), measuring the dimensions and describing the morphology of conidia (n = 50). At least 50 measurements were taken for taxonomically relevant structures when possible.

2.3. Genomic DNA Extraction, PCR Amplifications and Sequencing

From the isolates obtained, three representative strains (SZY21, SZY22, and SZY23) were selected for DNA extraction and sequencing because they were collected from different towns and years and showed typical *Colletotrichum* morphology with stable growth in culture. These isolates also produced lesions on *C. officinalis* leaves in pathogenicity assays and were therefore used as representative strains for molecular identification. Mycelial plugs (0.5 cm diameter) were excised from actively growing cultures on PDA and incubated for 5 days. Genomic DNA was extracted from approximately 200 mg (fresh weight) of mycelium using liquid nitrogen and the CTAB method [21].

Target loci were amplified using the primers in Table 1 (Sangon Biotech, Shanghai, China). Reactions (50 μL) contained 25 μL of Premix Taq (Ex Taq v2.0 plus dye; Takara RR902A), 1 μL of template DNA, 1 μL of each primer, and nuclease-free water to volume. Cycling: 98 °C for 5 min; 35 cycles of 95 °C for 30 s, 55 °C for 30 s, and 72 °C for 60 s; final extension, 72 °C for 10 min. Amplicons were verified on 0.8% agarose gels and Sanger-sequenced (Sangon Biotech, Shanghai, China).

Table 1. The primer sequences for each locus of *Colletotrichum* sp. in this study.

Locus	Primer	Sequences	References
ITS1-5.8S-ITS2	ITS1 ITS4	TCCGTAGGTGAACCTGCGG TCCTCCGCTTATTGATATGC	[22]
gapdh	GDF1 GDR1	GCCGTCAACGACCCCTTCATTGA GGGTGGAGTCGTACTTGAGCATGT	[23]
chs-1	CHS-79F CHS-354R	TGGGGCAAGGATGCTTGGAAGAAG TGGAAGAACCATCTGTGAGAGTTG	[24]
act	ACT-512F ACT-783R	ATGTGCAAGGCCGGTTTCGC TACGAGTCCTTCTGGCCCAT	[24]
tub2	T1 Bt2b	AACATGCGTGAGATTGTAAGT ACCCTCAGTGTAGTGACCCTTGGC	[25,26]

2.4. Phylogenetic Analyses

All sequence data generated in this study were deposited in GenBank. Newly obtained sequences were queried against the GenBank database using BLAST (version 2.17.0) (https://blast.ncbi.nlm.nih.gov/, accessed on 15 October 2025). The sequences of *Colletotrichum* spp. were retrieved from GenBank (Table 2). For each locus, our sequences, together with representative references retrieved from GenBank, were aligned with

MAFFT [27]. The resulting alignments were automatically trimmed with trimAl v1.5.0 and inspected, with manual adjustments made where necessary [28]. Bayesian inference (BI) was conducted on the concatenated dataset comprising the internal transcribed spacer (ITS1-5.8S-ITS2), glyceraldehyde-3-phosphate dehydrogenase (*gapdh*), chitin synthase (*chs-1*), actin (*act*), and β-tubulin (*tub2*). BI analyses were performed in MrBayes v3.2 [29], with the best-fitting substitution model for each partition selected by ModelFinder [30], and the workflow was managed in PhyloSuite v1.2.3 [31]. Bayesian posterior probabilities (BPPs) are shown at the corresponding nodes of a phylogenetic tree.

Table 2. Isolates and GenBank accession numbers used for phylogenetic analyses in this study.

Species and Strains	Host and Country	ITS1-5.8S-ITS2	*gapdh*	*chs-1*	*act*	*tub2*
C. arecacearum						
LC13850	*Arecaceae* sp. China	MZ595867	MZ664049	MZ799262	MZ664165	MZ673986
LC13851	*Arecaceae* sp. China	MZ595868	MZ664050	MZ799263	MZ664166	MZ673987
C. cordylinicola						
GUCC 12080	*Cordyline fruticose* China	OP722973	OP784100	OP730653	OP740195	OP761966
GUCC 12079	*Cordyline fruticose* China	OP722972	OP784099	OP730652	OP740194	OP761965
C. jiangxiense						
GUCC 12044	*Illicium simonsii* China	OP722970	OP784038	OP730595	OP740135	OP761906
GUCC 12043	*Illicium simonsii* China	OP722998	OP784037	OP730594	OP740134	OP761905
C. gardeniae						
GUCC 12049	*Gardenia jasminoides* China	OP722959	OP737963	OP715766	OP715801	OP720858
GUCC 12048	*Gardenia jasminoides* China	OP722989	OP737962	OP715765	OP715800	OP720857
C. kunmingense						
GUCC 12053	*Ophiopogon japonicus* China	OP722975	OP737965	OP715769	OP715804	OP720861
C. grevilleae						
CBS 132879	*Grevillea* sp. Italy	KC297078	KC297010	KC296987	KC296941	KC297102
C. ligustri						
GUCC 12111	*Ilex chinensis* China	OP722988	OP737968	OP715773	OP740216	OP720864
C. endophytica						
LC0324	*Pennisetum purpureum* Thailand	KC633854	KC832854	MZ799261	KF306258	MZ673954
C. fructicola						
CBS 130416	*Coffea arabica* Thailand	JX010165	JX010033	JX009866	FJ907426	JX010405
GUCC12102	*Curcuma phaeocaulis* China	OP723019	OP784115	OP730667	OP740203	OP761981
GUCC 12059	*Ligustrum lucidum* China	OP723017	OP784075	OP730628	OP740172	OP761941
ICMP 18646	*Tetragastris panamensis* Panama	JX010173	JX010032	JX009874	JX009581	JX010409

Table 2. *Cont.*

Species and Strains	Host and Country	ITS1-5.8S-ITS2	*gapdh*	*chs-1*	*act*	*tub2*
C. siamense						
GUCC 12062	*Alocasia macrorrhiza* China	OP722957	OP784086	OP730639	OP740181	OP761952
GUCC 12061	*Hymenocallis littoralis* China	OP722977	OP784083	OP730636	OP740178	OP761949
CBS 130417	*Coffea arabica* Thailand	JX010171	JX009924	JX009865	FJ907423	JX010404
NTCC 1308	*Afrocarpus gracilior* India	PX250322	PX257861	PX257860	PX257856	PX257859
HSPCS214	*Clausena lansium* China	PX096396	PX120585	PX120717	PX120655	PX120781
C4, SZY21, SZY22, SZY23	*Artocarpus heterophyllus* Thailand	PP960241	PP982581	PP982566	PP975273	PP982550
	Cornus officinalis China	PV529834	PV540228	PV540225	PV544191	PV540231
	Cornus officinalis China	PV529835	PV540229	PV540226	PV544192	PV540232
	Cornus officinalis China	PV529836	PV540230	PV540227	PV544193	PV540233

2.5. Pathogenicity Tests

Pathogenicity tests were conducted using two approaches. For detached-leaf inoculation, fresh, tender, healthy leaves of *C. officinalis* were collected, rinsed for approximately 10 min, and transferred to a laminar-flow cabinet. Sterile Petri dishes containing 3 to 5 layers of autoclaved filter paper were pre-moistened with sterile water. Leaves were gently punctured using a disposable syringe needle, agar plugs taken from the actively growing margins of colonies (mycelial side facing the leaf) were placed on the wound site, and the control leaves received sterile PDA plugs. Plates were sealed and incubated at 25 °C, and symptoms were monitored daily. Upon symptom development, the pathogen was re-isolated from lesion margins and morphologically compared with the original inoculum to verify identity. To compare virulence among isolates, disease severity was evaluated under identical inoculation and incubation conditions. Digital images were captured at 0, 2, 3 dpi. Lesion diameter (mm) and area (mm^2) were quantified from calibrated images (ImageJ version 1.54p). Virulence was determined by comparing mean lesion area among isolates; the isolate producing the largest lesions (significantly larger than the others, $p < 0.05$) was considered the most virulent.

For conidial-suspension inoculation, actively sporulating colonies were rinsed with sterile water to obtain a conidial suspension (10^5 conidia per ml). Ten healthy *C. officinalis* potted plants (two years old) were selected. Leaves were surface-sterilized and air-dried. Without prior wounding, the conidial suspension (5 mL) was evenly sprayed onto the prepared healthy leaves for each plant, while control plants received sterile water. Inoculated potted plants were incubated at 25 °C for 10 days with a relative humidity of 90%. Incubation period (first visible symptom) was recorded at 1, 2, 3, 5, 7, 10 dpi. The disease severity index (DSI) calculated according to a five-grade scale (0 = healthy, 1 = <5% leaf area affected, 2 = 6–10%, 3 = 11–25%, and 4 = >25%). DSI (%) = $\sum(ni \times si)/(N \times s_{max}) \times 100$, where n_i is the number of leaves in severity class s_i, N is the total number of leaves, and $s_{max} = 4$. Group differences were tested by one-way ANOVA. Following symptom expression, the pathogen was re-isolated from lesion margins and compared with the original inoculum to confirm identity.

2.6. In Vitro Sensitivity of C. siamense to Fungicides

A representative and highly virulent isolate of *C. siamense*, obtained from symptomatic *C. officinalis* leaves, was selected for fungicide susceptibility testing because it consistently produced the largest lesions in pathogenicity assays compared with the other tested iso-

lates. Four fungicides that are commonly used for the management of foliar diseases were evaluated: carbendazim, tebuconazole, prochloraz, and difenoconazole (Hansheng Biotechnology, Qingdao, China). Stock solutions of each fungicide were prepared with sterile distilled water and added to molten potato dextrose agar (PDA) that had been autoclaved and cooled to approximately 50 °C. For carbendazim, final concentrations in PDA were 0.01, 0.02, 0.04, 0.08, and 0.16 mg a.i. L^{-1}. For tebuconazole, final concentrations were 0.05, 0.10, 0.20, 0.40, and 0.80 mg a.i. L^{-1}. For prochloraz, final concentrations were 0.10, 0.20, 0.40, 0.80, and 1.60 mg a.i. L^{-1}. For difenoconazole, final concentrations were 0.04, 0.08, 0.16, 0.32, and 0.64 mg a.i. L^{-1}. PDA plates without fungicide served as the untreated control. When an organic solvent was used to dissolve fungicides, solvent-only PDA plates were included as additional controls.

Mycelial plugs (5 mm in diameter) were cut from the actively growing margin of 7-day-old *C. siamense* colonies on PDA and placed in the center of fungicide-amended or non-amended PDA plates with the mycelium side down. All plates were incubated at 25 °C in the dark for 5 days, or until the colony in the untreated control nearly reached the edge of the plate. Each fungicide concentration was replicated three times, and the experiment was repeated once, yielding similar results. At the end of the incubation period, colony diameter was measured along two perpendicular axes on each plate, and the mean colony diameter was calculated after subtracting the initial plug diameter. The percentage inhibition of mycelial growth (I, %) for each treatment was calculated using the following formula, $I\ (\%) = (C-T)/C \times 100$, where C is the mean colony diameter in the untreated control and T is the mean colony diameter in the fungicide treatment. For each fungicide, the effective concentration required to reduce mycelial growth by 50% (EC_{50}) was estimated by probit analysis of the relationship between percentage inhibition and the logarithm of fungicide concentration using SPSS version 31. Data were subjected to one-way analysis of variance (ANOVA), and treatment means were separated by Tukey's honestly significant difference (HSD) test at $p = 0.05$.

3. Results

3.1. Disease Survey and Symptom Description

During the summers of 2023 and 2024, field surveys were conducted in *C. officinalis* orchards located in Taiping Town and Miping Town, Xixia County. The total surveyed area encompassed approximately 50 mu, a traditional Chinese area unit, ≈3.33 ha. The outbreak in 2023 coincided with mean temperatures of 22.5–28.5 °C, relative humidity (RH) of 70–82%, and monthly rainfall of 89.5–213.0 mm during June–September. In 2024, conditions during June-September included 26.0–28.5 °C, RH 71–82%, and rainfall of 53.3–540.6 mm, which favor anthracnose infections via prolonged leaf wetness. In addition, dense canopy and limited pruning likely increased canopy humidity and inoculum retention. Disease incidence ranged from 50% to 60%. The mean DSI (%) values across survey plots ranged from 32.4% to 36.8%, indicating moderate to severe disease pressure. Accordingly, the risk of spread to other regions will depend on the overlap of suitable temperature–humidity windows, rainfall or irrigation patterns that sustain leaf wetness, and canopy management practices.

As an initial etiological screen, 80 randomly selected symptomatic leaf specimens were examined microscopically. *Colletotrichum*-like conidia were detected in 43/80 specimens. Among these 43 specimens, 8 also contained *Alternaria*-like conidia, suggesting possible mixed colonization on some lesions. The remaining 37/80 specimens did not show observable conidia, likely because the lesions were at an early stage with limited sporulation. These microscopy observations were therefore used as preliminary evidence to guide subsequent isolation.

Anthracnose, caused by *Colletotrichum* spp., was the predominant foliar disease observed in the region. Symptoms were primarily confined to the leaves, initially appearing near the apex within interveinal tissue between the two lateral veins. Early lesions were irregularly shaped, dark brown to blackish brown, and exhibited indistinct margins. As the disease advanced, lesions enlarged and coalesced, forming extensive necrotic areas. In severe cases, necrosis progressed from the leaf tip toward the base, leading to complete foliar blight and premature leaf death (Figure 1).

Figure 1. Symptoms of leaf necrosis on *C. officinalis* in the field. (**a**) Early stage disease symptoms; (**b**–**d**) late-stage symptoms. The red arrows in (**b**,**d**) indicate lesions that progress along the veins.

Quantitative lesion measurements revealed that individual necrotic lesions ranged from 2.1 ± 0.4 mm to 8.6 ± 1.2 mm in diameter during the early stages, with merged lesions covering more than 30% of the leaf area in advanced infections. Field observations indicated that heavily infected trees exhibited premature defoliation, reduced photosynthetic canopy, and a potential impact on fruit yield and quality.

3.2. Morphological Identification of Colletotrichum Isolates

During June–September of 2023 and 2024, symptomatic *C. officinalis* leaves were collected from orchards in Taiping Town and Miping Town, Xixia County, Henan Province, China. A total of 68 isolates were obtained, and the three representative strains from distinct orchards (SZY21, SZY22, SZY23) displayed typical *Colletotrichum* colony morphology (Table 3). Initially, colonies on potato dextrose agar (PDA) were white, with flocculent and dense aerial mycelia and irregular margins. With culture aging, some regions gradually developed a dark pigmentation (Figure 2a).

Table 3. Characterization for the three representative isolates from *C. officinalis* in this study.

Isolate	Year/Month	Location	Host Tissue
SZY21	2023.08	Taiping Town	Leaf lesion
SZY22	2023.08	Taiping Town	Leaf lesion
SZY23	2024.09	Miping Town	Leaf lesion

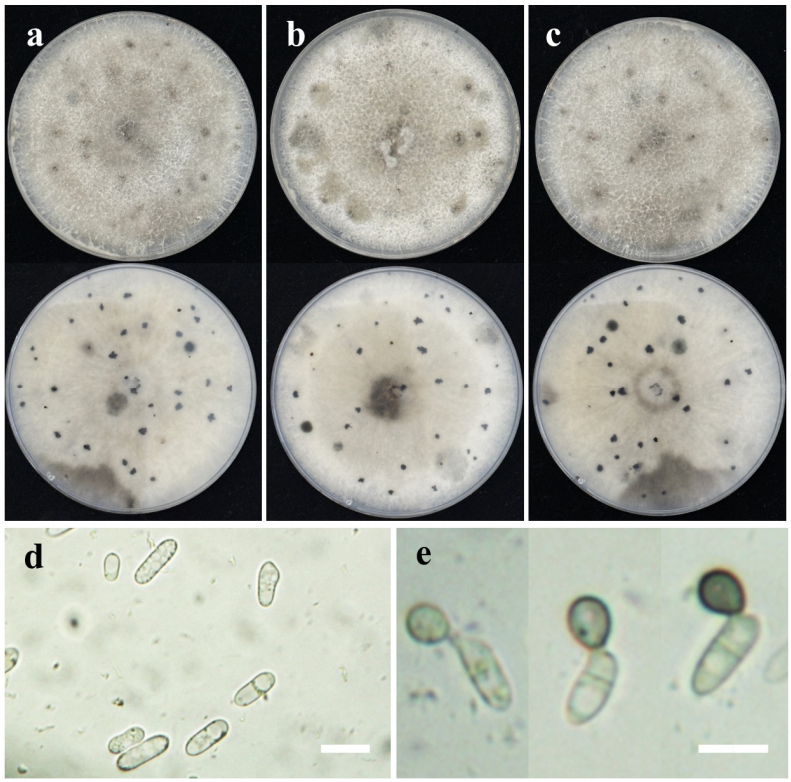

Figure 2. Morphology of the three representative isolates of SZY21, SZY22, and SZY23. (**a**–**c**) Colony morphology (obverse and reverse) on PDA after 21 days of incubation at 25 °C; (**a**) SZY21, (**b**) SZY22, and (**c**) SZY23; (**d**) Conidia. Scale bar = 20 µm. (**e**) Appressoria showing pronounced melanin accumulation. Scale bar = 20 µm.

Vegetative hyphae were 2–3.5 µm in diameter, hyaline, smooth-walled, septate, and branched. Conidia were hyaline, smooth-walled, aseptate, straight, and cylindrical, with both ends rounded, measuring 11–19 × 4–6 µm (mean ± SD = 15.1 ± 1.0 × 5.5 ± 0.4 µm), with a length-to-width (L/W) ratio of 2.7 (Figure 2d). Appressoria were abundant on slide cultures, medium to dark brown, single or in small clusters, clavate to broadly ellipsoidal, and sometimes irregularly lobed (Figure 2e). These morphological features were consistent with descriptions of *Colletotrichum siamense* within the *C. gloeosporioides* species complex [32].

Among the remaining isolates, 65/68 (95.6%) exhibited colony color, conidial size, and appressorial form consistent with *C. siamense*. A rapid marker screen on 20 additional isolates (ITS1-5.8S-ITS2) returned best BLAST matches to *C. siamense* (identity ≥ 99%, coverage ≥ 98%), with no conflicting top hits. These findings corroborate the predominance of *C. siamense* in the surveyed orchards.

3.3. Phylogenetic Analysis

Five genetic loci—ITS1-5.8S-ITS2, *gapdh*, *chs-1*, *act*, and *tub2*—were successfully amplified, sequenced, and deposited in GenBank under the accession numbers PV529834–PV529836 (ITS1-5.8S-ITS2), PV540228–PV540230 (*gapdh*), PV540225–PV540227 (*chs-1*), PV544191–PV544193 (*act*), and PV540231–PV540233 (*tub2*). Sequences from each locus

were aligned with those of phylogenetically related *Colletotrichum* species, as determined via BLAST searches (version 2.16.0) in the NCBI nucleotide database. Phylogenetic reconstruction based on the concatenated dataset of ITS1-5.8S-ITS2, *gapdh*, *chs-1*, *act*, and *tub2* generated a well-resolved tree (Figure 3). The alignment is composed of 1259 characters (ITS1-5.8S-ITS2: 1–470, *gapdh*: 471–607, *chs-1*: 608–818, *act*: 819–936, *tub2*: 937–1259). *Colletotrichum arecacearum* (LC13850 and LC13851) were used as outgroup taxa. Bayesian Inference phylogenies were inferred using MrBayes under K80 + I + G model (2 parallel runs, 2,000,000 generations), in which the initial 25% of sampled data were discarded as burn-in. The analyses consistently indicated that the strains obtained from *C. officinalis* represent a previously described species. The *C. officinalis* isolates clustered with *C. siamense* with high support, consistent with morphological evidence.

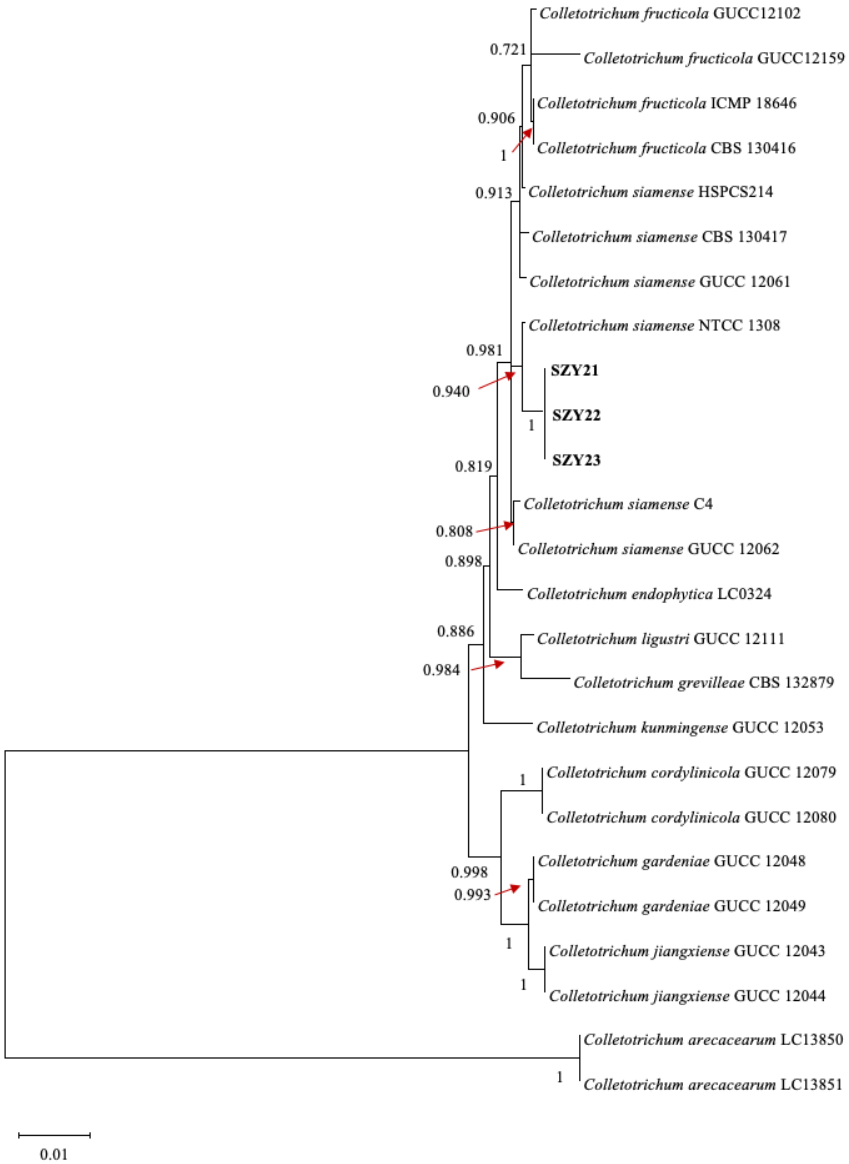

Figure 3. Phylogenetic trees based on Bayesian inference (BI) analysis of a combined dataset of ITS1-5.8S-ITS2, *gapdh*, *chs-1*, *act*, and *tub2* gene sequences of *Colletotrichum* species. Bayesian posterior probabilities (BPPs) are presented above the branches. Red arrows indicate the BPPs that were shifted slightly to avoid overlap with tree branches. Bold indicates the strains described in this study. The scale bar indicates 0.01 substitutions per site. Red arrows indicate the Bayesian posterior probability (BPP) values that were shifted slightly to avoid overlap with tree branches. *C. arecacearum* (LC13850 and LC13851) was selected as the outgroup.

3.4. Pathogenicity Test

Pathogenicity assays conducted using both detached-leaf and conidial-suspension inoculation approaches successfully reproduced typical anthracnose symptoms observed in the field. In the detached-leaf assay, necrotic lesions initiated at the inoculation sites within three days post-inoculation progressively enlarged into irregular dark brown to black spots, which often coalesced to form extensive blighted areas. No symptoms developed on control leaves inoculated with sterile PDA plugs. In the conidial-suspension assay, inoculated potted plants exhibited initial chlorotic spots within seven days, which subsequently expanded into irregular necrotic lesions consistent with field observations, with a disease incidence exceeding 90% by 10 days post-inoculation (Figure 4). Control plants sprayed with sterile water remained symptomless. *C. siamense* was consistently re-isolated from symptomatic tissues in both assays, and its morphological and molecular characteristics were identical to those of the original inoculum, thereby fulfilling Koch's postulates.

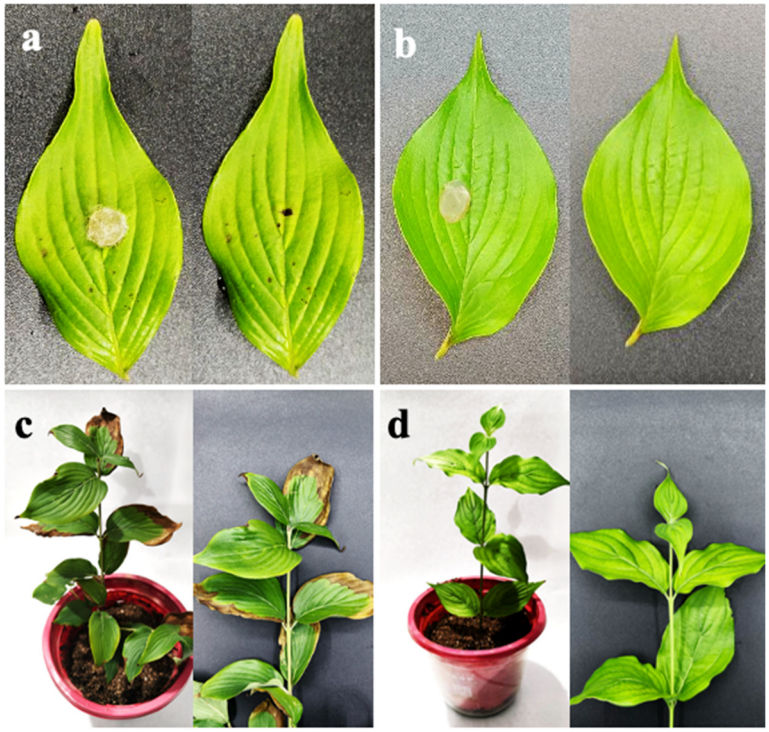

Figure 4. Pathogenicity assay results. (**a**) Symptoms on inoculated leaves; (**b**) no symptoms can be observed on the healthy control leaves; (**c**) symptoms on inoculated potted seedlings, 10 days after the inoculation with conidial suspension; (**d**) no symptoms on control plants.

Detached leaves developed necrotic spots by 2–3 dpi. Mean lesion diameter reached 1.8 ± 0.3 mm at 2 dpi and 3.2 ± 0.4 mm at 3 dpi (LSMeans ± SE); lesion area increased from 3.3 ± 0.2 mm^2 (2 dpi) to 9.1 ± 1.2 mm^2 (3 dpi) ($p < 0.001$ for time; treatment × time $p < 0.001$). In spray assays, the incubation period was 3.6 ± 0.2 days. DSI rose from 8.3 ± 1.1% (3 dpi) to 35.9 ± 2.5% (10 dpi). Among the tested isolates, SZY21 produced the largest lesions on inoculated leaves and was therefore considered the most virulent under our assay conditions; accordingly, SZY21 was used as the representative isolate for fungicide susceptibility testing. These assay values align with field observations (field mean DSI 32.4–36.8%) under warm, humid conditions. In detached-fruit inoculation assays, neither inoculated nor mock-treated harvested fruits developed visible lesions during the observation period.

3.5. Fungicide Sensitivity Assays

All four fungicides significantly inhibited mycelial growth of *C. siamense* on PDA compared with the untreated control ($p < 0.05$). For each fungicide, mycelial growth on PDA generally decreased with increasing fungicide concentration. Based on dose–response analyses, the EC_{50} values of carbendazim, tebuconazole, prochloraz, and difenoconazole against *C. siamense* were 0.035, 0.211, 0.386, and 0.391 mg a.i. L^{-1}, respectively (Table 4). Among the tested fungicides, carbendazim exhibited the lowest EC_{50} value, followed by tebuconazole and prochloraz, indicating high intrinsic activity against the pathogen. Difenoconazole showed a much higher EC_{50} value than the other fungicides, suggesting relatively weaker efficacy under the tested conditions. These results indicate that carbendazim, tebuconazole, and prochloraz are more effective options for inhibiting mycelial growth of *C. siamense* in vitro and may be considered as promising candidates for chemical management of leaf anthracnose on *C. officinalis*.

Table 4. Mycelial growth inhibition and EC_{50} estimates of *C. siamense* for four fungicides.

Fungicide	Tested Concentration Range (mg a.i. L^{-1})	Inhibition (%)	EC_{50} (mg a.i. L^{-1})
Carbendazim	0.01–0.16	43.22–58.26	0.035
Tebuconazole	0.05–0.80	42.91–72.40	0.211
Prochloraz	0.10–1.60	58.70–80.15	0.386
Difenoconazole	0.04–0.64	34.85–75.60	0.391

4. Discussion

Previous studies have documented several foliar diseases affecting *C. officinalis*, including powdery mildew (*Erysiphe* sp.), reported in Korea [33], and *Botryosphaeria dothidea* [20] and *Didymella* species [19], all of which primarily target the leaves rather than fruits. These pathogens are associated with reduced photosynthetic area and premature leaf senescence, ultimately impacting plant vigor and yield potential. In the present study, the pathogen isolated from symptomatic *C. officinalis* leaves was identified as *C. siamense*, representing the first record of this species causing anthracnose on this host. This finding indicates a novel disease occurrence in *C. officinalis* populations.

Our two-year field surveys indicate an epidemiological window for leaf anthracnose in *C. officinalis* orchards from June to September, characterized by warm temperatures (≈22.5–28.5 °C) together with recurrent rainfall leaf-wetness events and elevated canopy humidity. Such conditions are widely recognized as conducive to anthracnose epidemics and sustained surface wetness promote *Colletotrichum* conidial germination, appressorium formation, and secondary inoculum production. Consistently, studies have shown that conidial processes of *Colletotrichum* are strongly temperature–wetness dependent [34], and weather-based advisory systems commonly use temperature, leaf wetness duration, rainfall, and relative humidity to define infection-risk periods and optimize spray timing [35]. Moreover, *C. siamense* has been reported to grow best at 25–30 °C, which matches the observed summer window in Xixia County. Therefore, management should prioritize reducing leaf wetness, and we recommend weekly scouting from June to September and initiating interventions when disease levels begin to rise (DSI $\geq$ 10%) and when forecasts indicate prolonged wetness No fruit symptoms were observed in the field, and our postharvest inoculations did not produce lesions, suggesting limited fruit susceptibility under local microclimate. *Colletotrichum* fruit infections can be latent and become evident during storage [36]. Notably, no visible lesions developed on harvested fruits in our inoculation assays, and the mock controls remained symptomless as well. This negative result suggests that harvested *C. officinalis* fruits may be less permissive to anthracnose

symptom expression under the conditions tested. *Colletotrichum* infection typically begins with conidial adhesion and germination on the fruit surface, followed by appressorium-mediated penetration through the epidermis; therefore, fruit epidermal features can act as a physical barrier that restricts or delays colonization and symptom development [37].

The relatively low EC_{50} values of carbendazim, tebuconazole, and prochloraz against *C. siamense* in this study are comparable to those reported for this species on other hosts, suggesting that these fungicides could be integrated into management programs for leaf anthracnose on *C. officinalis*, while resistance risks and environmental safety should be carefully considered. In practice, chemical control should rely on foliar sprays applied at least one month prior to fruit harvest, to minimize fungicide residues on the harvested fruits and ensure the safe use of *C. officinalis* as a medicinal material.

Beyond *C. officinalis*, *C. siamense* has been documented on multiple pharmacologically important plants, underscoring its broad host range and epidemiological relevance in TCM systems [38]. Notable first reports include anthracnose on *Illicium verum* in China [39]. Similarly, *Kadsura coccinea*, a valued medicinal liana, has been shown to suffer from leaf anthracnose caused by *C. siamense* in Guangxi [40]. In woody ornamentals with medicinal or aromatic value, such as *Osmanthus fragrans*, it has been associated with high-incidence leaf anthracnose in South China [41]. Collectively, these cases demonstrate that *C. siamense* is both frequent and infects diverse medicinal plants, supporting our inference that it may warrant proactive monitoring and targeted management in herb medicine agroecosystems.

Species within the genus *Colletotrichum* can be differentiated based on colony morphology and conidial characteristics, including pigmentation, texture, growth rate, and the size and shape of conidia. Key microscopic traits, such as conidial length, width, and length-to-width ratio, as well as the presence and dimensions of appressoria, often serve as important diagnostic features. However, species-level identification based solely on morphological criteria is unreliable due to phenotypic plasticity and overlap in morphometric ranges among closely related taxa. Consequently, the current taxonomic consensus advocates the integration of multi-locus DNA sequence analyses for accurate phylogenetic delimitation of species within the genus. In the present study, five genetic loci—ITS1-5.8S-ITS2, *gapdh*, *chs-1*, *act*, and *tub2*—were sequenced and concatenated to construct a multi-locus phylogeny [42,43]. This approach is consistent with recent studies that have employed combined morphological and molecular datasets to resolve species boundaries within the *C. gloeosporioides* species complex. Representative species used for comparative analysis included *C. cordylinicola*, *C. endophytica*, *C. fructicola*, *C. gardeniae*, *C. grevilleae*, *C. jiangxiense*, *C. kunmingense*, *C. ligustri*, and *C. siamense*. Species delimitation within the *C. gloeosporioides* complex relies on multi-locus evidence benchmarked to ex-type references rather than morphology alone. In our dataset, the concatenated tree—rooted with *C. arecacearum* (LC13850, LC13851)—places the isolates within *C. siamense* with strong support (BI 0.940). Locus-specific BLAST results showed higher identity to *C. siamense* ex-type than to *C. fructicola* across informative markers, and pairwise p-distances to *C. siamense* were consistently smaller than to *C. fructicola*. For example, *gapdh* and *act*—which are particularly informative in this complex—displayed 100% (114/114) and 100% (126/126) of SZY21 identity to the *C. siamense* ex-type CBS 130417, respectively, versus 98% (112/114) and 99% (125/126) to *C. fructicola* ex-type CBS 130416. In the concatenated phylogeny rooted with *C. arecacearum* (LC13850/LC13851), our isolates (SZY21–SZY23) are nested within the *C. siamense* clade and cluster with *C. siamense* NTCC 1308. The relatively long terminal branch leading to our isolates likely reflects concatenation across informative loci and uneven within-species sampling, rather than placement outside *C. siamense*. Morphological traits are congruent with *C. siamense*, supporting this identification. Together, these lines of evidence support identification as *C. siamense*.

While multi-locus phylogeny was performed on three representatives, the morphological concordance across isolates and rapid marker identifications support *C. siamense* as the dominant pathogen during the outbreak. Nonetheless, low-frequency co-occurring taxa cannot be fully excluded; future work will expand multi-locus sequencing and/or culture-independent metabarcoding to improve detection sensitivity.

5. Conclusions

This study provides the first report of the isolation and identification of the causal agent of anthracnose on *C. officinalis* leaves. Through the integration of culture-based isolation, ITS1-5.8S-ITS2-region sequencing, and pathogenic assays, *C. siamense* was confirmed as the predominant pathogen associated with symptomatic leaves and was demonstrated to cause disease in artificial inoculations, thereby fulfilling Koch's postulates. In addition, in vitro fungicide sensitivity assays showed that *C. siamense* was most sensitive to carbendazim (lowest EC_{50} value; 0.035 mg a.i. L^{-1}), providing preliminary guidance for chemical control. Collectively, these findings provide a scientific basis for targeted management of leaf anthracnose on *C. officinalis* in Henan Province and support carbendazim as a promising candidate for chemical control pending field validation.

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